
Chapter One

Mathematical Preliminaries

- 1.0 Introduction
- 1.1 Basic Concepts and Taylor’s Theorem
- 1.2 Orders of Convergence and Additional Basic Concepts
- 1.3 Difference Equations

1.0 Introduction

This chapter starts with a review of some important topics in calculus that are required in the subsequent chapters. We encourage readers to skip boldly over material that is already familiar to them. In fact, some may wish to begin with Chapter 2, p. 37.

1.1 Basic Concepts and Taylor’s Theorem

We begin with a review of some basic concepts from calculus. At this point, one might ask: *Why do we need to discuss such topics if we are primarily interested in scientific computing and numerical algorithms?* A solid background in basic mathematical concepts is essential to understanding the derivation of most numerical algorithms. Taylor’s Theorem in various forms is fundamental to many numerical procedures and is an excellent starting point for the study of scientific computing since no advanced mathematical concepts are required.

Limit, Continuity, and Derivative

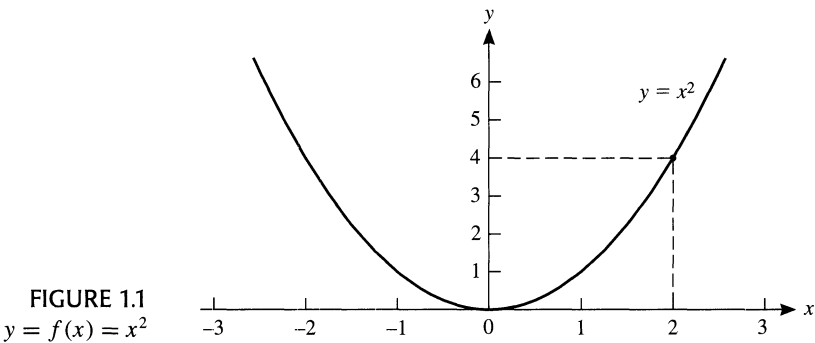
If f is a real-valued function of a real variable, then the **limit** of the function f at c (if it exists) is defined as follows: The equation

$$\lim_{x \rightarrow c} f(x) = L$$

means that to each positive ε there corresponds a positive δ such that the distance between $f(x)$ and L is less than ε whenever the distance between x and c is less than δ ; that is,

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad 0 < |x - c| < \delta$$

If there is no number L with this property, the limit of f at c does not exist.



For example, we consider the function

$$f(x) = x^2$$

From the graph of $f(x) = x^2$ in Figure 1.1, it appears that as x approaches 2, $f(x)$ approaches 4, so the equation

$$\lim_{x \rightarrow 2} x^2 = 4$$

should be true. We now see why this is so by proving that for $\varepsilon > 0$ there is a $\delta > 0$ such that $|x^2 - 4| < \varepsilon$ whenever $0 < |x - 2| < \delta$. Let $\varepsilon > 0$ and $\delta = -2 + \sqrt{4 + \varepsilon} > 0$ so that $\delta(\delta + 4) = \varepsilon$. If $0 < |x - 2| < \delta$, then $|x + 2| = |x - 2 + 4| \leq |x - 2| + 4 < \delta + 4$. Thus, we have $|x^2 - 4| = |x + 2||x - 2| < (\delta + 4)\delta = \varepsilon$. Notice that we worked backwards in a sense to discover what values of δ would give us exactly ε . Clearly, other values of δ work as well, such as $\delta = \varepsilon / (5 + \varepsilon)$. (See Problem 1.1.1, p. 12.)

As another example, we consider

$$g(x) = \frac{|x|}{x} = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$$

See Figure 1.2 for the graph of the function $g(x) = |x|/x$, from which it is clear why $g(x)$ is undefined at 0. We note that the equation

$$\lim_{x \rightarrow 0} \frac{|x|}{x} = L$$

is not true for *any* number L . Indeed, let $\varepsilon = 1$, and suppose that $||x|/x - L| < 1$ whenever $0 < |x| < \delta$. If $x = \frac{1}{2}\delta$, then $0 < |x| < \delta$ and we have $|x|/x = 1$. But if $x = -\frac{1}{2}\delta$, then $0 < |x| < \delta$ and $|x|/x = -1$. In both cases, we must have $||x|/x - L| < 1$. There is no number L satisfying both $|1 - L| < 1$ and $|-1 - L| < 1$, since this would require both $0 < L < 2$ and $-2 < L < 0$. Clearly, this is an impossible situation! So we say that the limit does *not* exist.

If f is defined only on a specified subset X of the real line, the definition of limit is modified so that $|f(x) - L| < \varepsilon$ whenever $x \in X$ and $0 < |x - c| < \delta$.

The function f is said to be **continuous** at c if

$$\lim_{x \rightarrow c} f(x) = f(c)$$

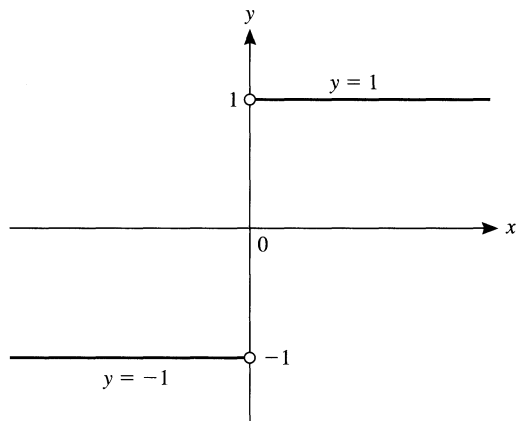


FIGURE 1.2
 $y = g(x) = |x|/x$

Thus, the function $f(x) = x^2$ is continuous at the point 2, whereas the function $|x|/x$ is not continuous at 0, no matter how it is defined at 0. These assertions follow from remarks made previously.

An intuitively obvious theorem is given next.

THEOREM 1 Intermediate-Value Theorem for Continuous Functions

On an interval $[a, b]$, a continuous function assumes all values between $f(a)$ and $f(b)$.

The **derivative** of f at c (if it exists) is defined by the equation

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

Since this limit need not exist for a particular function and a particular c , it is possible for the derivative not to exist for such a function. If f is a function for which $f'(c)$ exists, we say that f is **differentiable** at c . If f is differentiable at c , then f must be continuous at c . We now see why this is so. Consider

$$\begin{aligned} \lim_{x \rightarrow c} [f(x) - f(c)] &= \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} (x - c) \\ &= f'(c) \cdot \lim_{x \rightarrow c} (x - c) = f'(c) \cdot 0 = 0 \end{aligned}$$

Clearly, if $f(x)$ is differentiable at c , then $f'(x)$ exists and $\lim_{x \rightarrow c} f(x) = f(c)$.

But the converse is not true! For example, if

$$f(x) = |x|$$

then $f'(0)$ does not exist. See the graph of $f(x) = |x|$ in Figure 1.3. The derivative at a point x is the tangent line to the curve at $f(x)$. But at the bottom of the “V” in the curve ($x = 0$) there is no unique tangent and no derivative at $x = 0$.

The set of all functions that are continuous on the entire real line $\mathbb{R}$ is denoted by $C(\mathbb{R})$. The set of functions for which f' is continuous everywhere is denoted by $C^1(\mathbb{R})$. If $f \in C^1(\mathbb{R})$, then f' is continuous at all points in $\mathbb{R}$ and, thereby,

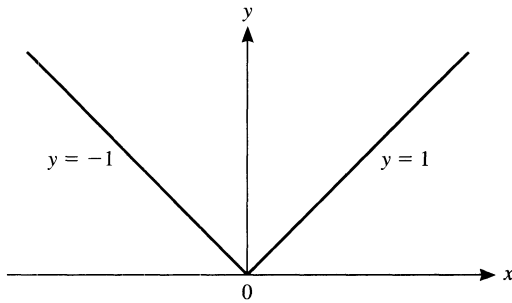


FIGURE 1.3
 $y = f(x) = |x|$

differentiable throughout the real line. Since the differentiability of a function at a point implies its continuity at that point, we have $C^1(\mathbb{R}) \subset C(\mathbb{R})$. The set $C^1(\mathbb{R})$ is a *proper* subset of $C(\mathbb{R})$ because there are (many) continuous functions whose derivatives do not exist. The function $f(x) = |x|$ is such an example.

We denote by $C^2(\mathbb{R})$ the set of all functions for which f'' is continuous everywhere. By reasoning similar to that given above,

$$C^2(\mathbb{R}) \subset C^1(\mathbb{R}) \subset C(\mathbb{R})$$

Again, these are proper inclusions since there are functions that are once differentiable but not twice, such as $f(x) = x^2 \sin(1/x)$. (See Problem 1.1.3, p. 12.)

Similarly, we define $C^n(\mathbb{R})$, for each natural number n , to be the set of all functions for which $f^{(n)}(x)$ is continuous. Finally, $C^\infty(\mathbb{R})$ is the set of functions each of whose derivatives is continuous. We have now

$$C^\infty(\mathbb{R}) \subset \cdots \subset C^2(\mathbb{R}) \subset C^1(\mathbb{R}) \subset C(\mathbb{R})$$

A familiar function in $C^\infty(\mathbb{R})$ is $f(x) = e^x$.

In the same way, we define $C^n[a, b]$ to be the set of functions f for which $f^{(n)}$ exists and is continuous on the closed interval $[a, b]$.

Taylor's Theorem

An important theorem concerning functions in $C^n[a, b]$ is Taylor's Theorem, which arises throughout the study of numerical analysis or in the study of numerical algorithms in scientific computing.

THEOREM 2 Taylor's Theorem with Lagrange Remainder

If $f \in C^n[a, b]$ and if $f^{(n+1)}$ exists on the open interval (a, b) , then for any points c and x in the closed interval $[a, b]$,

$$f(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(c)(x - c)^k + E_n(x) \tag{1}$$

where, for some point ξ between c and x , the error term is

$$E_n(x) = \frac{1}{(n + 1)!} f^{(n+1)}(\xi)(x - c)^{n+1}$$

Here “ ξ between c and x ” means that either $c < \xi < x$ or $x < \xi < c$ depending on the particular values of x and c involved.

An important special case arises when $c = 0$. Equation (1) becomes the **Maclaurin series** for $f(x)$:

$$f(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(0)x^k + E_n(x) \quad (2)$$

where

$$E_n(x) = \frac{1}{(n+1)!} f^{(n+1)}(\xi)x^{n+1}$$

We can obtain Taylor series for many important functions such as

$$\sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} \quad (-\infty < x < \infty)$$

$$\cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} \quad (-\infty < x < \infty)$$

$$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} x^k \quad (-1 < |x| < \infty)$$

$$\frac{1}{1+x} = \sum_{k=0}^{\infty} (-1)^k x^k \quad (-1 < x < 1)$$

The series that occur in these illustrations are **power series**. (See Section 6.7, p. 388.)

EXAMPLE 1 Determine the Taylor series of the following function, using Taylor's Theorem,

$$f(x) = \ln x$$

taking $a = 1$, $b = 2$, and $c = 1$.

Solution The formula requires various derivatives of $f(x) = \ln x$, which are $f'(x) = x^{-1}$, $f''(x) = -x^{-2}$, $f'''(x) = 2x^{-3}$, $f^{(4)}(x) = -6x^{-4}$, and so on. Next, we obtain the general term*

$$f^{(k)}(x) = (-1)^{k-1} (k-1)! x^{-k} \quad (k \geq 1)$$

Clearly, at $x = 1$, we have

$$f^{(k)}(1) = (-1)^{k-1} (k-1)! \quad (k \geq 1)$$

*Throughout when we use inequalities involving integer values such as $1 \leq i \leq m$, we mean $i = 1, 2, \dots, m$, and similarly, $n \geq N$ means $n = N, N+1, \dots$

Of course, $f^{(0)}(1) = f(x) = \ln 1 = 0$. Putting all of this into Taylor's formula (1) gives us

$$\ln x = \sum_{k=1}^n (-1)^{k-1} \frac{1}{k} (x-1)^k + E_n(x) \quad (1 \leq x \leq 2)$$

where

$$E_n(x) = (-1)^n \frac{1}{n+1} \xi^{-(n+1)} (x-1)^{n+1} \quad (1 < \xi < x) \quad \blacksquare$$

In the equation given for $\ln x$, the summation $\sum_{k=1}^n$ on the right of the equal sign produces a *polynomial* function in x as shown below. It can be regarded as a simple function that approximates the more complicated function $\ln x$. The last term on the right, $E_n(x)$, can be regarded as an error term. It tells us how the polynomial approximation differs from $\ln x$. Note that this term is *not* a polynomial because ξ depends on x in a nonpolynomial way.

We can use Taylor's formula for computing approximations of functions at particular points. For example, writing out the formula for $\ln x$, we have

$$\ln x = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \cdots + (-1)^{n-1} \frac{1}{n}(x-1)^n + E_n(x)$$

where

$$|E_n(x)| = \frac{1}{n+1} \xi^{-(n+1)} (x-1)^{n+1} < \frac{1}{n+1} (x-1)^{n+1}$$

In this estimate, we simply note that $1 < \xi$ and $\xi^{-(n+1)} < 1$.

EXAMPLE 2 How many terms in the series need to be used to compute $\ln 2$ with accuracy of one part in 10^8 ?

Solution Letting $x = 2$, we have then

$$\begin{aligned} \ln 2 &= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + (-1)^{n-1} \frac{1}{n} + E_n(2) \\ &= \sum_{k=1}^n (-1)^{k-1} \frac{1}{k} + E_n(2) \end{aligned}$$

with $|E_n(2)| < 1/(n+1)$. The term $E_n(2)$ is the numerical error. So to compute $\ln 2$ with the desired accuracy would require n to be chosen so that $E_n(2) \leq 10^{-8}$. This means $1/(n+1) \leq 10^{-8}$ or $n+1 \geq 10^8$. Thus, at least *100 million* terms in the polynomial would be required to compute $\ln 2$ to the desired accuracy! We conclude that using Taylor series is not a practical way to compute $\ln 2$, and some other procedure needs to be used. In fact, a similar calculation shows that $\ln 1.5$ can be computed to the same accuracy with only 22 terms! (See Problem 1.1.5, p. 13.) $\blacksquare$

The special case $n = 0$ of Taylor's Theorem is often used in mathematical arguments. It is known as the **Mean-Value Theorem**.

THEOREM 3 Mean-Value Theorem

If f is in $C[a, b]$ and if f' exists on the open interval (a, b) , then for x and c in the closed interval $[a, b]$,

$$f(x) = f(c) + f'(\xi)(x - c)$$

where ξ is between c and x .

Taking $x = b$ and $c = a$ and rearranging, we have the important equation

$$f(b) - f(a) = f'(\xi)(b - a) \quad \text{where} \quad a < \xi < b$$

An approximation to $f'(x)$ can be obtained from the Mean-Value Theorem:

$$f'(x) \approx \frac{f(x + h) - f(x - h)}{2h}$$

This is discussed in Section 7.1 (p. 465).

A special case of the Mean-Value Theorem is **Rolle's Theorem**.

THEOREM 4 Rolle's Theorem

If f is continuous on $[a, b]$, if f' exists on (a, b) , and if $f(a) = f(b)$, then $f'(\xi) = 0$ for some ξ in the open interval (a, b) .

This is an immediate consequence of an equation above. (Actually, in a formal development, Rolle's Theorem is proved first, and from it, Taylor's Theorem is derived.) In both Rolle's Theorem and the Mean-Value Theorem, there may be more than one point ξ in the interval $[a, b]$ that satisfies the given equation.

In Section 7.6 (p. 515), we shall need another form of Taylor's Theorem as given next. It has several nice features—the proof is straightforward and Taylor's Theorem with Lagrange Remainder can be obtained directly from Taylor's Theorem with Integral Remainder.

THEOREM 5 Taylor's Theorem with Integral Remainder

If $f \in C^{n+1}[a, b]$, then for any points x and c in the closed interval $[a, b]$,

$$f(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(c)(x - c)^k + R_n(x) \quad (3)$$

where

$$R_n(x) = \frac{1}{n!} \int_c^x f^{(n+1)}(t)(x - t)^n dt$$

Proof Recall the formula for integration by parts

$$\int u \, dv = uv - \int v \, du$$

and apply it to the integral R_n with

$$u = \frac{(x-t)^n}{n!} \quad dv = f^{(n+1)}(t) \, dt$$

The result is

$$\begin{aligned} R_n &= \frac{1}{n!} \left[f^{(n)}(t)(x-t)^n \Big|_{t=c}^{t=x} + n \int_c^x f^{(n)}(t)(x-t)^{n-1} \, dt \right] \\ &= -\frac{1}{n!} f^{(n)}(c)(x-c)^n + R_{n-1} \end{aligned}$$

If this process of integrating by parts is repeated, we eventually obtain

$$R_n = -\sum_{k=1}^n \frac{1}{k!} f^{(k)}(c)(x-c)^k + R_0$$

Since

$$R_0 = \int_c^x f'(t) \, dt = f(x) - f(c)$$

we have

$$f(x) = f(c) + \sum_{k=1}^n \frac{1}{k!} f^{(k)}(c)(x-c)^k + R_n$$

and this completes the proof. ■

Other Forms of Taylor's Formula

Yet another form of the series and the error term in Taylor's formula can be obtained by replacing x by $x+h$ and c by x in Taylor's Theorem with Lagrange Remainder. The result follows.

THEOREM 6 Alternative Form of Taylor's Theorem

If $f \in C^{n+1}[a, b]$, then for any points x and $x+h$ in the closed interval $[a, b]$,

$$f(x+h) = \sum_{k=0}^n \frac{h^k}{k!} f^{(k)}(x) + E_n(h) \tag{4}$$

where

$$E_n(h) = \frac{h^{n+1}}{(n+1)!} f^{(n+1)}(\xi)$$

in which the point ξ lies between x and $x+h$.

In detail, (4) is

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{3!}f'''(x) + \cdots + \frac{h^n}{n!}f^{(n)}(x) + E_n(h)$$

It is an important form for many applications.

EXAMPLE 3 Determine Taylor's formula for A^{x+h} and approximate $10^{1.0001}$.

Solution Letting $f(x) = A^x$, we find that $f^{(n)}(x) = A^x(\ln A)^n$. Using Equation (4), we have

$$A^{x+h} = A^x \left(1 + \sum_{k=1}^n \frac{h^k}{k!} (\ln A)^k \right) + E_n(h)$$

Next let $A = 10$, $x = 1$, and $h = 10^{-4}$. Thus, we obtain

$$\begin{aligned} 10^{1.0001} &= 10 \left(1 + 10^{-4}(\ln 10) + \frac{1}{2}10^{-8}(\ln 10)^2 + \cdots \right) \\ &\approx 10(1 + 2.30259 \times 10^{-4} + 2.65095 \times 10^{-8}) \\ &\approx 10.0023000265095 \end{aligned}$$

■

For vector-valued functions of vectors, there also exist Taylor series and Taylor formulas. Thus, if f is a mapping of $\mathbb{R}^n$ to $\mathbb{R}^m$, formulas are available for expressing $f(x+h)$ in terms of $f(x)$, $f'(x)$, $f''(x)$, and so on. Of course, the main difficulty lies in defining the appropriate derivatives. These matters are taken up in many textbooks, such as Bartle [1976], K.T. Smith [1971], and Dieudonné [1960]. Next we shall mention some special cases needed in later chapters.

For a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, the simplest expression of Taylor's formula is a symbolic one:

THEOREM 7 Taylor's Theorem in Two Variables

Let $f \in C^{n+1}([a, b] \times [c, d])$. If (x, y) and $(x+h, y+k)$ are points in the rectangle $[a, b] \times [c, d] \subseteq \mathbb{R}^2$, then

$$f(x+h, y+k) = \sum_{i=0}^n \frac{1}{i!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^i f(x, y) + E_n(h, k) \tag{5}$$

where

$$E_n(h, k) = \frac{1}{(n+1)!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^{n+1} f(x+\theta h, y+\theta k)$$

in which θ lies between 0 and 1.

The meaning of the mysterious terms in this theorem is as follows:

$$\begin{aligned}\left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^0 f(x, y) &= f(x, y) \\ \left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^1 f(x, y) &= \left(h\frac{\partial f}{\partial x} + k\frac{\partial f}{\partial y}\right)(x, y) \\ \left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^2 f(x, y) &= \left(h^2\frac{\partial^2 f}{\partial x^2} + 2hk\frac{\partial^2 f}{\partial x \partial y} + k^2\frac{\partial^2 f}{\partial y^2}\right)(x, y)\end{aligned}$$

and so on. Letting $f_x = \partial f/\partial x$, $f_y = \partial f/\partial y$, $f_{xx} = \partial^2 f/\partial x^2$, $f_{xy} = \partial^2 f/\partial x \partial y$, $f_{yy} = \partial^2 f/\partial y^2$, we can write the first few terms of (5) as

$$f(x + h, y + k) = f + (hf_x + kf_y) + \frac{1}{2}(h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}) + \cdots$$

where on the right-hand side the function f and each of the following partial derivatives are evaluated at (x, y) .

EXAMPLE 4 What are the first few terms in the Taylor formula for $f(x, y) = \cos(xy)$?

Solution For the given function, we find that

$$\begin{aligned}\frac{\partial f}{\partial x} &= -y \sin(xy) & \frac{\partial f}{\partial y} &= -x \sin(xy) \\ \frac{\partial^2 f}{\partial x^2} &= -y^2 \cos(xy) & \frac{\partial^2 f}{\partial x \partial y} &= -xy \cos(xy) - \sin(xy) & \frac{\partial^2 f}{\partial y^2} &= -x^2 \cos(xy)\end{aligned}$$

Thus, if we let $n = 1$ in Taylor’s formula (5), the result is

$$\cos[(x + h)(y + k)] = \cos(xy) - hy \sin(xy) - kx \sin(xy) + E_1(h, k)$$

The remainder E_1 is the sum of three terms—namely,

$$\begin{aligned}& -\frac{1}{2}h^2(y + \theta k)^2 \cos[(x + \theta h)(y + \theta k)] \\ & -hk\{(x + \theta h)(y + \theta k) \cos[(x + \theta h)(y + \theta k)] + \sin[(x + \theta h)(y + \theta k)]\} \\ & -\frac{1}{2}k^2(x + \theta h)^2 \cos[(x + \theta h)(y + \theta k)]\end{aligned}\quad \blacksquare$$

PROBLEMS 1.1

1. Show that $|x^2 - 4| < \varepsilon$ when $0 < |x - 2| < \varepsilon(5 + \varepsilon)^{-1}$ and prove $\lim_{x \rightarrow 2} x^2 = 4$ by using these inequalities.
2. Show that the function $f(x) = x \sin(1/x)$, with $f(0) = 0$, is continuous at 0 but not differentiable at 0.
3. Show that $f(x) = x^2 \sin(1/x)$, with $f(0) = 0$, is once differentiable at 0 but not twice.

4. Let $f(x) = x^{-3}(x - \sin x)$ for $x \neq 0$. How should $f(0)$ be defined in order that f be continuous? Will it also be differentiable?
5. a. Derive the Taylor series at 0 for the function $f(x) = \ln(x + 1)$. Write this series in summation notation. Give two expressions for the remainder when the series is truncated.
- b. Determine the smallest number of terms that must be taken in the series to yield $\ln 1.5$ with an error less than 10^{-8} .
- c. Determine the number of terms necessary to compute $\ln 1.6$ with error 10^{-10} at most.
6. Determine whether the following function is continuous, once differentiable, or twice differentiable:

$$f(x) = \begin{cases} x^3 + x - 1 & \text{if } x \leq 0 \\ x^3 - x - 1 & \text{if } x > 0 \end{cases}$$

7. (Continuation) Repeat the preceding problem for the function

$$f(x) = \begin{cases} x & \text{if } x \leq 1 \\ x^2 & \text{if } x > 1 \end{cases}$$

8. Criticize this reasoning: The function f defined by

$$f(x) = \begin{cases} x^3 + x & \text{if } x \leq 0 \\ x^3 - x & \text{if } x \geq 0 \end{cases}$$

has the properties

$$\lim_{x \rightarrow 0^+} f''(x) = \lim_{x \rightarrow 0^+} 6x = 0$$

$$\lim_{x \rightarrow 0^-} f''(x) = \lim_{x \rightarrow 0^-} 6x = 0$$

Therefore, f'' is continuous.

9. Prove that if f is differentiable at x , then

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x-h)}{2h} = f'(x)$$

Show that for some functions that are not differentiable at x , the preceding limit exists. (See Eggermont [1988] or the following problem.)

10. Prove or disprove this assertion: If f is differentiable at x , then for $\alpha \neq 1$,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x+\alpha h)}{h - \alpha h}$$

11. Show that $\lim_{x \rightarrow 1} (4x + 2) = 6$ by means of an ε - δ proof.
12. Show that $\lim_{x \rightarrow 2} (1/x) = \frac{1}{2}$ by means of an ε - δ proof.
13. For the function $f(x) = 3 - 2x + x^2$ and the interval $[a, b] = [1, 3]$, find the number ξ that occurs in the Mean-Value Theorem.
14. (Continuation) Repeat the preceding problem with the function $f(x) = x^6 + x^4 - 1$ and the interval $[0, 1]$.
15. Find the Taylor series for $f(x) = \cosh x$ about the point $c = 0$.
16. If the series for $\ln x$ is truncated after the term involving $(x - 1)^{1000}$ and is then used to compute $\ln 2$, what bound on the error can be given?

17. Find the Taylor series for $f(x) = e^x$ about the point $c = 3$. Then simplify the series and show how it could have been obtained directly from the series for f about $c = 0$.
18. Let k be a positive integer and let $0 < \alpha < 1$. To what classes $C^n(\mathbb{R})$ does the function $x^{k+\alpha}$ belong?
19. Prove: If $f \in C^n(\mathbb{R})$, then $f' \in C^{n-1}(\mathbb{R})$ and $\int_a^x f(t) dt$ belong to $C^{n+1}(\mathbb{R})$.
20. Prove Rolle's Theorem directly (not as a special case of the Mean-Value Theorem).
21. Prove: If $f \in C^n(\mathbb{R})$ and $f(x_0) = f(x_1) = \cdots = f(x_n) = 0$ for $x_0 < x_1 < \cdots < x_n$, then $f^{(n)}(\xi) = 0$ for some $\xi \in (x_0, x_n)$. *Hint:* Use Rolle's Theorem n times.
22. Prove that the function $f(x) = x^2$ is continuous everywhere.
23. For small values of x , the approximation $\sin x \approx x$ is often used. Estimate the error in using this formula with the aid of Taylor's Theorem. For what range of values of x will this approximation give results correct to six decimal places?
24. For small values of x , how good is the approximation $\cos x \approx 1 - \frac{1}{2}x^2$? For what range of values will this approximation give correct results rounded to three decimal places?
25. Use Taylor's Theorem with $n = 2$ to prove that the inequality $1 + x < e^x$ is valid for all real numbers except $x = 0$.
26. Derive the Taylor series with remainder term for $\ln(1 + x)$ about 1. Derive an inequality that gives the number of terms that must be taken to yield $\ln 4$ with error less than 2^{-m} .
27. What is the third term in the Taylor expansion of $x^2 + x - 2$ about the point 3?
28. Using the series for e^x , how many terms are needed to compute e^2 correctly to four decimal places (rounded)?
29. Develop the Taylor series for $f(x) = \ln x$ about e , writing the results in summation notation and giving the remainder term. Suppose $|x - e| < 1$ and accuracy $\frac{1}{2} \times 10^{-1}$ is desired. What is the minimum number of terms in the series required to achieve this accuracy?
30. Determine the first two terms of the Taylor series for x^x about 1 and the remainder term E_1 .
31. Determine the Taylor polynomial of degree 2 for $f(x) = e^{(\cos x)}$ expanded about the point π .
32. First develop the function $\sqrt{x}$ in a series of powers of $(x - 1)$ and then use it to approximate $\sqrt{0.9999999995}$ to ten decimal places.
33. Assume that $|x| < \frac{1}{2}$ and determine by Taylor's Theorem the best upper bound.
 - a. $|\cos x - (1 - x^2/2)|$
 - b. $|\sin x - x(1 - x^2/6)|$
34. Determine a function that can be termed the **linearization** of $x^3 - 2x$ at 2.
35. How many terms are required in the series

$$e = \sum_{k=0}^{\infty} \frac{1}{k!}$$

to give e with an error of at most 6/10 unit in the 20th decimal place?

36. Find the first two terms in the Taylor expansion of $x^{1/5}$ about the point $x = 32$. Approximate the fifth root of 31.999999 using these two terms in the series. How accurate is your answer?
37. Find the Taylor polynomial of degree 2 for the function $f(x) = e^{2x} \sin x$ expanded about the point $\pi/2$.

- 38. Determine the Lagrange form of the remainder when Taylor’s Theorem is applied to the function $f(x) = \cos x$, with $n = 2$ and $c = \pi/2$. How small must we make $|x - \pi/2|$ if this remainder term is not to exceed $\frac{1}{2} \times 10^{-4}$ in absolute value?
- 39. An error term of the form $(-1)^n(n + 1)^{-1}\xi^{-n-1}(x - 1)^{n+1}$ was obtained in the example illustrating Taylor’s formula. Compare this to the error term that arises from the integral form of the remainder.
- 40. Use Taylor’s Theorem with Integral Remainder and the Mean-Value Theorem for Integrals to deduce Taylor’s Theorem with Lagrange Remainder.

1.2 Orders of Convergence and Additional Basic Concepts

In numerical calculations, especially on high-performance computers, it often happens that the answer to a problem is not produced all at once. Rather, a sequence of approximate answers is produced, usually exhibiting progressively higher accuracy. Convergence of sequences is an important subject that will be taken up again later, such as in Chapter 3 (p. 73). Here we present just a few introductory concepts.

Convergent Sequences

Let us consider an idealized situation in which a single real number is sought as the answer to a problem. It might be, for example, a zero of a complicated equation or the numerical value of an intractable definite integral. In such a case, a computer program may produce a sequence of real numbers $x_1, x_2, x_3, \dots$ that are *approaching* the correct answer.

We write

$$\lim_{n \rightarrow \infty} x_n = L$$

if there corresponds to each positive ε a real number r such that $|x_n - L| < \varepsilon$ whenever $n > r$. (Here n is an integer.)

For example,

$$\lim_{n \rightarrow \infty} \frac{n + 1}{n} = 1$$

because

$$\left| \frac{n + 1}{n} - 1 \right| < \varepsilon$$

whenever $n > \varepsilon^{-1}$.

For another example, recall the equation

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n$$

by which the important irrational number e can be defined. If we compute the sequence $x_n = (1 + 1/n)^n$, some of the elements are

$$\begin{aligned}x_1 &= 2.00000\ 0 \\x_{10} &= 2.59374\ 2 \\x_{30} &= 2.67431\ 9 \\x_{50} &= 2.69158\ 8 \\x_{1000} &= 2.71692\ 4\end{aligned}$$

This is an example of a sequence that is converging rather slowly, since the limit is $e = 2.71828\ 18\ \dots$ and in the 1000th term there is still an error of 0.00135 8. Using double-precision computations, we find numerical evidence that

$$\frac{|x_{n+1} - e|}{|x_n - e|} \rightarrow 1$$

This property is worse than **linear convergence** (to be defined precisely later).
An example of a sequence that converges to 0 at a slightly faster rate is

$$x_{n+1} = x_n - \frac{x_n^2}{x_n^2 + x_{n-1}^2}$$

Selecting two initial values, we have

$$\begin{aligned}x_0 &= 20.00 \\x_1 &= 15.00 \\x_2 &= 14.64 \\x_3 &= 14.15 \\x_{33} &= 0.54 \\x_{34} &= 0.27\end{aligned}$$

While faster than the previous example, the convergence is still quite slow. Using double-precision computations, we find numerical evidence that

$$\frac{|x_{n+1}|}{|x_n|} \rightarrow 0$$

which is called **superlinear convergence**.

As an example of a rapidly convergent sequence, consider the one defined recursively by putting

$$\begin{cases} x_1 = 2 \\ x_{n+1} = \frac{1}{2}x_n + \frac{1}{x_n} \end{cases} \quad (n \geq 1)$$

The elements of this sequence are

$$x_1 = 2.000000\ 0$$

$$x_2 = 1.500000\ 0$$

$$x_3 = 1.41666\ 7$$

$$x_4 = 1.41421\ 6$$

The limit is $\sqrt{2} = 1.41421\ 3562\dots$, and the sequence is converging to its limit with great rapidity. Using double-precision computations, we find evidence that

$$\frac{|x_{n+1} - \sqrt{2}|}{|x_n - \sqrt{2}|^2} \leq 0.36$$

Such a condition corresponds to **quadratic convergence** as we shall shortly see.

Orders of Convergence

Some special terminology is used to describe the rapidity with which a sequence converges. Let $[x_n]$ be a sequence of real numbers tending to a limit x^* . We say that the rate of convergence is at least **linear** if there are a constant $c < 1$ and an integer N such that

$$|x_{n+1} - x^*| \leq c|x_n - x^*| \quad (n \geq N)$$

We say that the rate of convergence is at least **superlinear** if there exist a sequence ε_n tending to 0 and an integer N such that

$$|x_{n+1} - x^*| \leq \varepsilon_n |x_n - x^*| \quad (n \geq N)$$

The convergence is at least **quadratic** if there are a constant C (not necessarily less than 1) and an integer N such that

$$|x_{n+1} - x^*| \leq C|x_n - x^*|^2 \quad (n \geq N)$$

In general, if there are positive constants C and α and an integer N such that

$$|x_{n+1} - x^*| \leq C|x_n - x^*|^\alpha \quad (n \geq N)$$

we say that the rate of convergence is of **order** α at least.

Big $\mathcal{O}$ and Little $\mathcal{o}$ Notation

We shall now consider several standard ways of comparing two sequences or two functions. We begin with sequences.

Let $[x_n]$ and $[\alpha_n]$ be two different sequences. We write

$$x_n = \mathcal{O}(\alpha_n)$$

if there are constants C and n_0 such that $|x_n| \leq C|\alpha_n|$ when $n \geq n_0$. Here we say that x_n is “big oh” of α_n .

The equation

$$x_n = \mathcal{O}(\alpha_n)$$

means intuitively that $\lim_{n \rightarrow \infty} (x_n/\alpha_n) = 0$. Here we say that x_n is “little oh” of α_n . To avoid division by zero, the careful definition must be that for some $\varepsilon_n \geq 0$ we have $\varepsilon_n \rightarrow 0$ and $|x_n| \leq \varepsilon_n |\alpha_n|$.

These two notions give a coarse method of comparing two sequences. They are frequently used when both sequences converge to 0. If $x_n \rightarrow 0$, $\alpha_n \rightarrow 0$, and $x_n = \mathcal{O}(\alpha_n)$, then x_n converges to 0 at least as rapidly as α_n does. If $x_n = \mathcal{O}(\alpha_n)$, then x_n converges to 0 more rapidly than α_n does.

Here are some examples:

$$\frac{n+1}{n^2} = \mathcal{O}\left(\frac{1}{n}\right) \tag{1}$$

$$\frac{1}{n \ln n} = \mathcal{O}\left(\frac{1}{n}\right) \tag{2}$$

$$\frac{1}{n} = \mathcal{O}\left(\frac{1}{\ln n}\right) \tag{3}$$

$$\frac{5}{n} + e^{-n} = \mathcal{O}\left(\frac{1}{n}\right) \tag{4}$$

$$e^{-n} = \mathcal{O}\left(\frac{1}{n^2}\right) \tag{5}$$

Taking an example from Section 1.1 (p. 8), we can write

$$\ln 2 - \sum_{k=1}^{n-1} (-1)^{k-1} \frac{1}{k} = \mathcal{O}\left(\frac{1}{n}\right) \tag{6}$$

This is an illustration of very **slow convergence**. On the other hand,

$$e^x - \sum_{k=0}^{n-1} \frac{1}{k!} x^k = \mathcal{O}\left(\frac{1}{n!}\right) \quad (|x| \leq 1) \tag{7}$$

and this illustrates very **rapid convergence**.

The notation just introduced is used also for functions other than sequences. For example, we can write

$$\sin x = x - \frac{x^3}{6} + \mathcal{O}(x^5) \quad (x \rightarrow 0) \tag{8}$$

This means that there exist a neighborhood of 0 and a constant C such that on that neighborhood

$$\left| \sin x - x + \frac{x^3}{6} \right| \leq C |x^5|$$

The correctness of this assertion can be verified by using Taylor's Theorem with $n = 4$ and $f(x) = \sin x$.

An equation of the form

$$f(x) = \mathcal{O}(g(x)) \quad (x \rightarrow \infty)$$

means that constants r and C exist so that $|f(x)| \leq C|g(x)|$ whenever $x \geq r$. Thus, for example,

$$\sqrt{x^2 + 1} = \mathcal{O}(x) \quad (x \rightarrow \infty)$$

since $\sqrt{x^2 + 1} \leq 2x$ when $x \geq 1$. In using the notation $f(x) = \mathcal{O}(g(x))$ or $f(x) = \mathcal{o}(g(x))$ it is important that we state what point of convergence is intended. For example, $x^{-2} = \mathcal{O}(x^{-1})$ as $x \rightarrow \infty$, but the relation is reversed at 0: $x^{-1} = \mathcal{O}(x^{-2})$ as $x \rightarrow 0$.

In general, we write

$$f(x) = \mathcal{O}(g(x)) \quad (x \rightarrow x^*)$$

when there is a constant C and a neighborhood of x^* such that $|f(x)| \leq C|g(x)|$ in that neighborhood. Similarly,

$$f(x) = \mathcal{o}(g(x)) \quad (x \rightarrow x^*)$$

means that $\lim_{x \rightarrow x^*} [f(x)/g(x)] = 0$.

Mean-Value Theorem for Integrals

Another mean-value theorem often needed in numerical analysis follows.

THEOREM 1 Mean-Value Theorem for Integrals

Let u and v be continuous real-valued functions on an interval $[a, b]$, and suppose that $v \geq 0$. Then there exists a point ξ in $[a, b]$ such that

$$\int_a^b u(x)v(x) dx = u(\xi) \int_a^b v(x) dx$$

Proof Let α and β denote the least and greatest values of $u(x)$ on $[a, b]$, respectively. Then

$$\alpha \leq u(x) \leq \beta \quad (a \leq x \leq b)$$

Since $v(x) \geq 0$, we have

$$\alpha v(x) \leq u(x)v(x) \leq \beta v(x)$$

Now integrate throughout this inequality and put $I = \int_a^b v(x) dx$. The result is

$$\alpha I \leq \int_a^b u(x)v(x) dx \leq \beta I$$

If $I = 0$, we can conclude that $v(x) \equiv 0$, and the result we wish to prove is trivial. If I is not 0, we have

$$\alpha \leq I^{-1} \int_a^b u(x)v(x) dx \leq \beta$$

By the Intermediate-Value Theorem for Continuous Functions, a point ξ exists in $[a, b]$ for which

$$u(\xi) = I^{-1} \int_a^b u(x)v(x) dx \quad \blacksquare$$

Nested Multiplication

An elementary concept that is needed in various places throughout the text is nested multiplication of polynomials. A polynomial can be rewritten in a nested form so that it requires only a few more than the minimum number of multiplications when evaluating it. The polynomial

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$$

can be written using standard mathematical notation involving the sum $\sum$ and product $\prod$ as follows:

$$p(x) = \sum_{k=0}^n a_k x^k = \sum_{k=0}^n \left(a_k \prod_{j=1}^k x \right)$$

Recall that if $n \leq m$, then

$$\sum_{k=n}^m s_k = s_n + s_{n+1} + \cdots + s_m \quad \text{and} \quad \prod_{k=n}^m y_k = y_n y_{n+1} \cdots y_m$$

and, thereby,

$$\sum_{k=n}^m r = (m - n + 1)r \quad \text{and} \quad \prod_{k=n}^m x = x^{(m-n+1)}$$

By convention, if $m < n$, then

$$\sum_{k=n}^m s_k = 0 \quad \text{and} \quad \prod_{k=n}^m y_k = 1$$

To evaluate the polynomial efficiently, we can group the terms using **nested multiplication**:

$$p(x) = a_0 + x(a_1 + x(a_2 + \cdots + x(a_{n-1} + x(a_n)) \cdots))$$

This corresponds to the following simple algorithm involving only n multiplications and n additions:

```
p ← an
for k = n - 1 to 0 step - 1 do
    p ← xp + ak
end do
```

This procedure is also known as **Horner’s method** or **synthetic division**.

Upper and Lower Bounds

Two important concepts that frequently arise in numerical analysis are **supremum (least upper bound)** and **infimum (greatest lower bound)**. Let S be a nonempty, bounded set of real numbers. **Boundedness** is the property that for appropriate real numbers a and b ,

$$a \leq x \leq b \qquad \text{for all } x \in S$$

In this situation, we call b an upper bound for S and a a lower bound for S . Of course, S has many upper bounds: If $c \geq b$, then c is also an upper bound for S . Obviously, there is no *largest* upper bound, but there is a *smallest* upper bound for the set S .

AXIOM 1 Least Upper Bound Axiom

Any nonempty set of real numbers that possesses an upper bound has a least upper bound.

The property expressed in the axiom is one of the deeper characteristics of the real number system. (The rational number system does not have this property, for example.) The **least upper bound** of a set S (if it has one) is denoted by $\text{lub } S$. The word **supremum** is synonymous with this and is abbreviated by $\sup S$. Thus, we can write

DEFINITION 1 Definition of Supremum

The supremum of S is v ($v = \sup S = \text{lub } S$) if and only if

- 1. *v is an upper bound for S and*
- 2. *no real number smaller than v is an upper bound for S .*

EXAMPLE 1 What is $\sup S$ if $S = \{x : x^2 < 2\}$?

Solution Since $S = \{x : -\sqrt{2} < x < \sqrt{2}\}$, the least upper bound is $\sqrt{2}$. ■

If F is a function, the notation $\sup_{x \in A} f(x)$ means $\sup\{f(x) : x \in A\}$. For example, we can show that

$$\sup_{0 < x < \frac{\pi}{6}} \sin x = \frac{1}{2}$$

The corresponding concept of **greatest lower bound** or **infimum** is written $u = \text{glb } S$ or $u = \inf S$ and is defined as follows:

DEFINITION 2 Definition of Infimum

The infimum of S is u ($u = \inf S = \text{glb } S$) if and only if

- 1. u is a lower bound for S and
- 2. no real number greater than u is a lower bound for S .

Here the basic result is that if a nonempty set of real numbers has a lower bound, then it has a greatest lower bound. This follows from the axiom when we apply it to the set $-S = \{-x : x \in S\}$.

Explicit and Implicit Functions

A function is usually defined via an explicit formula, from which a value of the function can be computed for each argument. For example, we can define a function f by the formula

$$f(x) = \sqrt{7x^3 - 2x}$$

After enlarging our repertoire of familiar functions, we can define more complicated functions such as

$$f(x) = \ln(\arctan x) + \cos(e^x)$$

There are, however, many other methods of defining functions, such as by means of a differential equation, an integral, or an infinite series. Thus, a function $y = f(x)$ is well defined by the following differential equation with an initial condition:

$$\begin{cases} y' = 1 + \sin y \\ y(0) = 0 \end{cases}$$

Another example, called the **error function** and denoted by $\text{erf}(x)$, is defined by an integral:

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

In this section, we consider functions defined **implicitly**. This means that a function G of *two* variables is given and, from the equation $G(x, y) = 0$, we hope to recover y as a function of x . In some cases, we can unravel the equation $G(x, y) = 0$ to obtain $y = f(x)$. For example, in the equation

$$y^2 + 3xy - 7 = 0$$

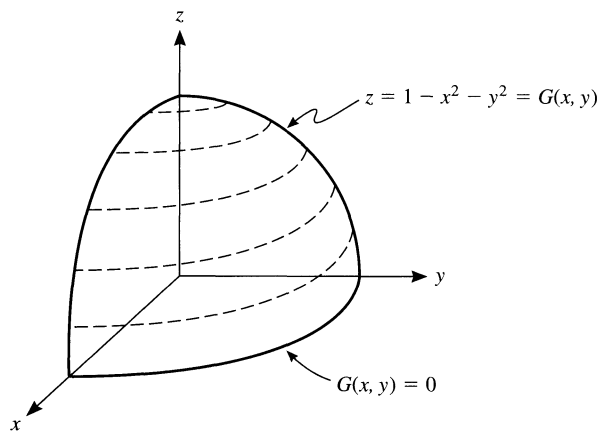
we can solve for y and get

$$y = \frac{1}{2}(-3x \pm \sqrt{9x^2 + 28})$$

thereby recovering *two* explicit functions. Likewise, from the equation

$$\sin(y + 7) = x^3 - 2$$

FIGURE 1.4
Surface G intersects
 xy -plane showing
curve $G(x, y) = 0$



we can recover an explicit function by solving for y :

$$y = \sin^{-1}(x^3 - 2) - 7$$

Here too there are multiple functions because various **branches** of the inverse sine function can be chosen.

In general, if G is a prescribed function, and if (x_0, y_0) is a particular point where $G(x_0, y_0) = 0$, then we expect that there will be additional nearby points that satisfy the equation $G(x, y) = 0$. Thus, when x is altered, we expect that a suitable change in y will be necessary to keep the equation $G(x, y) = 0$ true. Hence, y should be a function of x in some neighborhood of x_0 . Figure 1.4 shows a typical case.

The following important theorem governs this situation.

THEOREM 2 **Implicit Function Theorem**

Let G be a function of two real variables defined and continuously differentiable in a neighborhood of (x_0, y_0) . If $G(x_0, y_0) = 0$ and $\partial G/\partial y \neq 0$ at (x_0, y_0) , then there is a positive δ and a continuously differentiable function f defined for $|x - x_0| < \delta$ such that $f(x_0) = y_0$ and $G(x, f(x)) = 0$ for $|x - x_0| < \delta$.

EXAMPLE 2 Does the equation

$$x^7 + 2y^8 - y^3 = 0$$

define y as a continuously differentiable function of x in some neighborhood of $x = -1$?

Solution To answer this, let $(x_0, y_0) = (-1, 1)$ and put

$$G(x, y) = x^7 + 2y^8 - y^3$$

Then $G(x_0, y_0) = 0$ and

$$\frac{\partial G}{\partial y} = 16y^7 - 3y^2$$

Hence, $\partial G(x_0, y_0)/\partial y = 13$. By the Implicit Function Theorem, y is a continuously differentiable function of x near $x_0 = -1$. ■

If f is a function defined implicitly by $G(x, y) = 0$, then for x in some interval, $G(x, f(x)) = 0$. If we desire to compute $f'(x)$, a familiar technique from calculus can be used. We differentiate in the equation $G(x, y) = 0$ with respect to x , remembering that y is interpreted as a function of x . The result is

$$\frac{\partial G}{\partial x} + \frac{\partial G}{\partial y} \frac{dy}{dx} = 0$$

from which we obtain

$$\frac{dy}{dx} = -\frac{\partial G}{\partial x} / \frac{\partial G}{\partial y}$$

or

$$f'(x) = -\frac{\partial G}{\partial x} / \frac{\partial G}{\partial y}$$

The derivative thus obtained is likely to be a function of x and y .

EXAMPLE 3 What is the value of dy/dx at the point $(2, 1)$, if $y(x)$ is defined implicitly by the equation $x^3 - y^7 + 4x^2 + y^4 - 24 = 0$?

Solution Total differentiation with respect to x produces

$$3x^2 - 7y^6 \frac{dy}{dx} + 8x + 4y^3 \frac{dy}{dx} = 0$$

Substituting $x = 2$ and $y = 1$, we get

$$12 - 7 \frac{dy}{dx} + 16 + 4 \frac{dy}{dx} = 0$$

from which $dy/dx = 28/3$. ■

Some numerical problems involving implicit functions are discussed in Section 3.2 (p. 86).

PROBLEMS 1.2

1. Determine the best integer value of k in the equation

$$\arctan x = x + \mathcal{O}(x^k) \text{ as } x \rightarrow 0$$

2. Let a sequence x_n be defined inductively by $x_{n+1} = F(x_n)$. Suppose that $x_n \rightarrow x$ as $n \rightarrow \infty$ and $F'(x) = 0$. Show that

$$x_{n+2} - x_{n+1} = \mathcal{O}(x_{n+1} - x_n)$$

Hint: Use the Mean-Value Theorem and assume that F is a continuously differentiable function.

3. Prove that every sufficiently smooth function can be approximated on an interval of length h by a polynomial of degree n with an error that is $\mathcal{O}(h^{n+1})$ as $h \rightarrow 0$.

4. a. Consider the series

$$e^{\tan x} = 1 + x + \frac{x^2}{2!} + \frac{3x^3}{3!} + \frac{9x^4}{4!} + \cdots \quad (|x| \leq \pi/2)$$

Retaining three terms in the series, estimate the remaining series using $\mathcal{O}$ -notation with the best integer value possible, as $x \rightarrow 0$.

- b. Repeat the problem using $\mathcal{O}$ -notation and the series

$$\ln \tan x = \ln x + \frac{x^2}{3} + \frac{7x^4}{90} + \frac{62x^6}{2835} + \cdots \quad (0 < |x| < \pi/2)$$

5. Establish the range of integer values of γ and δ for which the $+\cdots$ in the series

$$\ln(1+x) = \sum_{k=1}^{n-1} (-1)^{k-1} \frac{x^k}{k} + \cdots$$

can be replaced with either $\mathcal{O}(x^\gamma)$ or $\mathcal{O}(x^\delta)$ as $x \rightarrow 0$.

6. For the pair (x_n, α_n) , is it true that $x_n = \mathcal{O}(\alpha_n)$ as $n \rightarrow \infty$?

a. $x_n = 5n^2 + 9n^3 + 1, \quad \alpha_n = n^2$

b. $x_n = 5n^2 + 9n^3 + 1, \quad \alpha_n = 1$

c. $x_n = \sqrt{n+3}, \quad \alpha_n = 1$

d. $x_n = 5n^2 + 9n^3 + 1, \quad \alpha_n = n^3$

e. $x_n = \sqrt{n+3}, \quad \alpha_n = 1/n$

7. Choose the correct assertions (in each, $n \rightarrow \infty$).

a. $(n+1)/n^2 = \mathcal{O}(1/n)$

b. $(n+1)/\sqrt{n} = \mathcal{O}(1)$

c. $1/\ln n = \mathcal{O}(1/n)$

d. $1/(n \ln n) = \mathcal{O}(1/n)$

e. $e^n/n^5 = \mathcal{O}(1/n)$

8. The expressions e^h , $(1-h^4)^{-1}$, $\cos(h)$, and $1 + \sin(h^3)$ all have the same limit as $h \rightarrow 0$. Express each in the following form with the best integer values of α and β .

$$f(h) = c + \mathcal{O}(h^\alpha) = c + \mathcal{O}(h^\beta)$$

9. (Continuation) What are the limit and the rate of convergence of the following expression as $h \rightarrow 0$?

$$\frac{1}{h^2} [(1+h) - e^h]$$

Express the limit in the form given in the preceding problem.

10. Show that these assertions are not true.

a. $e^x - 1 = \mathcal{O}(x^2)$ as $x \rightarrow 0$

b. $x^{-2} = \mathcal{O}(\cot x)$ as $x \rightarrow 0$

c. $\cot x = \mathcal{O}(x^{-1})$ as $x \rightarrow 0$

11. Let $[a_n] \rightarrow 0$ and let $\lambda > 1$. Show that $\sum_{k=0}^n a_k \lambda^k = \mathcal{O}(\lambda^n)$ as $n \rightarrow \infty$.

12. Explain why the least upper bound axiom does not apply to the empty set.

13. Find two functions in explicit form that are defined implicitly by the equation

$$(x^3 - 1)y + e^x y^2 + \cos x - 1 = 0$$

14. In solving differential equations, one often obtains solutions in implicit form. Show that the equation

$$2x^3y^2 + x^2y + e^x = c$$

defines a solution of the differential equation

$$\frac{dy}{dx} = -(6x^2y^2 + 2xy + e^x)/(4x^3y + x^2)$$

15. Kepler's equation in astronomy is $x - y + \varepsilon \sin y = 0$, where ε is a parameter in the range $0 \leq \varepsilon \leq 1$. Show that for each real x there is a real y that makes the equation true. Show that if $0 \leq \varepsilon < 1$, then dy/dx is continuous everywhere. *Hint:* Write $x = y - \varepsilon \sin y$ and consider the behavior of $y - \varepsilon \sin y$ as $y \rightarrow +\infty$ and as $y \rightarrow -\infty$. Use the Implicit Function Theorem for the second part of this problem.

16. Find the points x at which the equation

$$y - \ln(x + y) = 0$$

defines y implicitly as a function of x . Compute dy/dx .

17. Give an example to show why the least upper bound axiom does not apply to the set of all rational numbers.
18. Does the least upper bound axiom apply to the set of all integers?
19. What are the values of the following?

- a. $\sup_{x \in \mathbb{R}} \arctan x$
- b. $\sup_{x \geq 0} e^{-x}$
- c. $\inf_{x \in \mathbb{R}} e^{-x}$
- d. $\sup_{x \in \mathbb{R}} (x^2 + 1)^{-1}$

20. Use the Mean-Value Theorem for Integrals to prove that

$$\int_0^{\pi/2} e^x \cos x \, dx = e^y$$

for some y in $(0, \pi/2)$.

21. Give an example to show why the continuity of u cannot be dropped from the hypotheses in Theorem 1.
22. Prove that if $0 < \theta < 1$, then $(1 + a\theta^n)/(1 + a\theta^{n-1})$ converges to 1 linearly.
23. Are the following equivalent?
- a. $|f(x)| = \mathcal{O}(|x|^{-n-\varepsilon})$ for some $\varepsilon > 0$ as $|x| \rightarrow \infty$
 - b. $|f(x)| = \mathcal{O}(|x|^{-n})$
24. Prove that the set of upper bounds for a set S in $\mathbb{R}$ is either $\mathbb{R}$, the empty set, or an interval of the form $[a, \infty)$.
25. Prove by induction that the Horner algorithm is correct.
26. When the sequence $x_n = (1 + 1/n)^n$ is computed, it appears to be monotone increasing. Prove that this is so. *Hints:* First, if $\ln f(x)$ is increasing, then so is $f(x)$. Second, if $f'(x) > 0$, then f is increasing. Finally, $\ln x$ is defined to be $\int_1^x t^{-1} dt$.
27. (Continuation) Show that the elements of the sequence in the preceding problem remain less than 3.
28. Prove that $x_n = x + \mathcal{O}(1)$ if and only if $\lim_{n \rightarrow \infty} x_n = x$.

- 29. Verify the correctness of each expression in Equation (1) in the text.
- 30. For fixed n , show that

$$\sum_{k=0}^n x^k = 1/(1-x) + \mathcal{O}(x^n) \text{ as } x \rightarrow 0$$

- 31. Determine the best integer value for β such that for fixed n ,

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \mathcal{O}(x^\beta) \quad (0 < x < 1)$$

as $x \rightarrow 0$. Repeat for $\mathcal{O}(x^\beta)$. Is there a best integer value in this case? Explain.

- 32. Show that if $x_n = \mathcal{O}(\alpha_n)$, then $cx_n = \mathcal{O}(\alpha_n)$.
- 33. Show that if $x_n = \mathcal{O}(\alpha_n)$, then $x_n/\ln n = \mathcal{O}(\alpha_n)$.
- 34. Find the best integer value of k in the assertion $\cos x - 1 + x^2/2 = \mathcal{O}(x^k)$ as $x \rightarrow 0$.
- 35. Prove: If $x_n = \mathcal{O}(\alpha_n)$, then $x_n = \mathcal{O}(\alpha_n)$. Show that the converse is not true.
- 36. Prove: If $x_n = \mathcal{O}(\alpha_n)$ and $y_n = \mathcal{O}(\alpha_n)$, then $x_n + y_n = \mathcal{O}(\alpha_n)$.
- 37. Prove: If $x_n = \mathcal{O}(\alpha_n)$ and $y_n = \mathcal{O}(\alpha_n)$, then $x_n + y_n = \mathcal{O}(\alpha_n)$.
- 38. Show that for any $r > 0$, $x^r = \mathcal{O}(e^x)$ as $x \rightarrow \infty$.
- 39. Show that for any $r > 0$, $\ln x = \mathcal{O}(x^r)$ as $x \rightarrow \infty$.
- 40. Prove: If $\alpha_n \rightarrow 0$, $x_n = \mathcal{O}(\alpha_n)$, and $y_n = \mathcal{O}(\alpha_n)$, then $x_n y_n = \mathcal{O}(\alpha_n)$.
- 41. Prove: If $x_n = \mathcal{O}(\alpha_n)$, then $\alpha_n^{-1} = \mathcal{O}(x_n^{-1})$. Show that the same is true for the $\mathcal{O}$ -relationship.

COMPUTER
PROBLEMS 1.2

- 1. Consider the recursive relation

$$\begin{cases} x_0 = 1 & x_1 = c \\ x_{n+1} = x_n + x_{n-1} & (n \geq 1) \end{cases}$$

- a. It can be shown that when $c = (1 + \sqrt{5})/2$, a closed-form formula is given by

$$x_n = \left(\frac{1 + \sqrt{5}}{2}\right)^n$$

- b. Similarly, when $c = (1 - \sqrt{5})/2$,

$$x_n = \left(\frac{1 - \sqrt{5}}{2}\right)^n$$

- c. If $c = 1$, then

$$x_n = \frac{1}{\sqrt{5}} \left(\frac{1 + \sqrt{5}}{2}\right)^{n+1} - \frac{1}{\sqrt{5}} \left(\frac{1 - \sqrt{5}}{2}\right)^{n+1}$$

For all values of n in the range $1 \leq n \leq 30$, compute x_n by both the recursive relation and the formula for each case. Explain the results. This recursion defines the famous **Fibonacci sequence**.

- 2. Using a symbolic manipulation program, find the Taylor series about 0, up to and including the term x^{10} , for the function $(\tan x)^2$. Express the missing terms in $\mathcal{O}$ -notation.
- 3. Establish Equations (1) through (5) (see p. 18).

1.3 Difference Equations

Algorithms for numerical computation are often designed to produce *sequences* of numbers. It is useful, therefore, to develop a little of the theory of linear sequence spaces. This theory will be needed in Chapter 8 (p. 524) for the analysis of linear multistep methods. (We present it here because it requires little mathematical background to understand and thus lends itself to an introductory section that stands alone.)

Basic Concepts

In the following discussion, V will stand for the set of all infinite sequences of complex numbers, such as

$$\begin{aligned}x &= [x_1, x_2, x_3, \dots] \\ y &= [y_1, y_2, y_3, \dots]\end{aligned}$$

A sequence is, formally, a complex-valued function defined on the set of positive integers $\mathbb{N} = \{1, 2, 3, \dots\}$. It is simply a matter of convenience that we write x_n in place of $x(n)$ for the value of the function x at the argument n .

In the set V , we define two operations:

$$\begin{aligned}x + y &= [x_1 + y_1, x_2 + y_2, x_3 + y_3, \dots] \\ \lambda x &= [\lambda x_1, \lambda x_2, \lambda x_3, \dots]\end{aligned}$$

A more compact way of writing these equations is

$$\begin{aligned}(x + y)_n &= x_n + y_n \\ (\lambda x)_n &= \lambda x_n\end{aligned}$$

There is a 0 element in V : namely, $0 = [0, 0, 0, \dots]$. With the adoption of these definitions, V becomes a vector space. The verification of the vector-space postulates is tedious and not very instructive. The vector space V is infinite dimensional; indeed, the following set of vectors is linearly independent:

$$\begin{aligned}v^{(1)} &= [1, 0, 0, 0, \dots] \\ v^{(2)} &= [0, 1, 0, 0, \dots] \\ v^{(3)} &= [0, 0, 1, 0, \dots] \\ v^{(4)} &= [0, 0, 0, 1, \dots] \\ &\vdots\end{aligned}$$

We shall be interested in linear operators $L : V \rightarrow V$. One of the most important of these is the **shift operator** or **displacement operator**, denoted by E and defined by the equation

$$Ex = [x_2, x_3, x_4, \dots] \quad \text{where} \quad x = [x_1, x_2, x_3, \dots]$$

Thus,

$$(Ex)_n = x_{n+1}$$

It is clear that E can be applied several times in succession to produce, for example,

$$(EEx)_n = x_{n+2}$$

or

$$(E^kx)_n = x_{n+k}$$

In the remainder of this section, our attention will be confined to linear operators that can be expressed as linear combinations of powers of E . Such an operator is termed a **linear difference operator** (with constant coefficients and finite rank). The general form of such an operator is

$$L = \sum_{i=0}^m c_i E^i \tag{1}$$

Of course, E^0 is defined as the identity operator,

$$(E^0x)_n = (Ix)_n = x_n$$

From Equation (1), we see that the linear difference operators of the form (1) make up a linear subspace in the set of all linear operators from V into V . The powers of E provide a basis for this subspace.

Notice that L in Equation (1) is a *polynomial* in E ; in other words, it is a linear combination of powers of E . Thus, we could write

$$L = p(E)$$

where p is a polynomial called the **characteristic polynomial** of L and defined by

$$p(\lambda) = \sum_{i=0}^m c_i \lambda^i$$

The problem to be studied here is the determination of all the solutions to an equation of the form $Lx = 0$, where L is an operator of the type (1). From the linearity of L , it follows at once that the set $\{x : Lx = 0\}$ is a linear subspace of V ; it is called the **null space** of L . We can regard the equation $Lx = 0$ as being solved if a basis has been found for the null space of L .

To see what to expect in general, let us consider a concrete example of L , say by taking $c_0 = 2$, $c_1 = -3$, $c_2 = 1$, and all other $c_i = 0$. The resulting equation, which is known as a **linear difference equation**, can be written in three forms:

$$\begin{aligned} (E^2 - 3E^1 + 2E^0)x &= 0 \\ x_{n+2} - 3x_{n+1} + 2x_n &= 0 \quad (n \geq 1) \\ p(E)x &= 0 \quad p(\lambda) = \lambda^2 - 3\lambda + 2 \end{aligned} \tag{2}$$

It is very easy to generate sequences that solve (2). Indeed, we can choose x_1 and x_2 arbitrarily and then determine $x_3, x_4, \dots$ by use of (2). For instance, we can obtain in this way the various solutions

$$\begin{aligned} &[1, 0, -2, -6, -14, -30, \dots] \\ &[1, 1, 1, 1, \dots] \\ &[2, 4, 8, 16, \dots] \end{aligned}$$

The first solution is more mysterious than the second two because it is not clear at first glance what its general term is. The second two solutions are obviously of the form $x_n = \lambda^n$, with $\lambda = 1$ or 2 . It is natural to inquire whether any other such solutions exist. Putting $x_n = \lambda^n$ in (2) yields

$$\begin{aligned} \lambda^{n+2} - 3\lambda^{n+1} + 2\lambda^n &= 0 \\ \lambda^n(\lambda^2 - 3\lambda + 2) &= 0 \\ \lambda^n(\lambda - 1)(\lambda - 2) &= 0 \end{aligned}$$

This simple analysis shows that there is just one other solution of the type sought—namely, $[0, 0, 0, \dots]$. This we call the **trivial solution**. Now, it happens that the solutions u defined by $u_n = 1$ and v defined by $v_n = 2^n$ form a basis for the solution space of (2). To prove this, let x be **any solution** of (2). We seek constants α and β such that $x = \alpha u + \beta v$. This equation means that $x_n = \alpha u_n + \beta v_n$ for all n . In particular, for $n = 1$ and 2 , we have

$$\begin{cases} x_1 = \alpha + 2\beta \\ x_2 = \alpha + 4\beta \end{cases} \tag{3}$$

Equation (3) uniquely determines α and β because the determinant of the matrix

$$\begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix}$$

is not 0. (It is the determinant of a **Vandermonde matrix** as defined in Section 6.1, p. 314.) But now we can prove by induction that $x_n = \alpha u_n + \beta v_n$ for *all* n . Indeed, if this equation is true for indices less than n , then it is true for n because

$$\begin{aligned} x_n &= 3x_{n-1} - 2x_{n-2} \\ &= 3(\alpha u_{n-1} + \beta v_{n-1}) - 2(\alpha u_{n-2} + \beta v_{n-2}) \\ &= \alpha(3u_{n-1} - 2u_{n-2}) + \beta(3v_{n-1} - 2v_{n-2}) \\ &= \alpha u_n + \beta v_n \end{aligned}$$

This example illustrates the case of simple roots[†] of the characteristic polynomial.

[†]We should use the term **roots** when it is a polynomial being considered and **zero** for other (more general) functions. In higher-level mathematics, people always talk about the zeros of either a function or a polynomial since roots is in accordance with older usage.

Simple Roots

THEOREM 1 Theorem on Null Space

If p is a polynomial and λ is a root of p , then one solution of the difference equation $p(E)x = 0$ is $[\lambda, \lambda^2, \lambda^3, \dots]$. If all the roots of p are simple and nonzero, then each solution of the difference equation is a linear combination of such special solutions.

Proof First, if λ is any complex number and $u = [\lambda, \lambda^2, \lambda^3, \dots]$, then $Eu = \lambda u$ because

$$(Eu)_n = u_{n+1} = \lambda^{n+1} = \lambda(\lambda^n) = \lambda u_n$$

By reapplying the operator E , one obtains in general $E^i u = \lambda^i u$. Since E^0 has been defined as the identity map, we have $E^0 u = \lambda^0 u$. Thus, if p is a polynomial defined by $p(\lambda) = \sum_{i=0}^m c_i \lambda^i$, then

$$p(E)u = \left(\sum_{i=0}^m c_i E^i\right)u = \sum_{i=0}^m c_i (E^i u) = \sum_{i=0}^m c_i \lambda^i u = p(\lambda)u$$

If $p(\lambda) = 0$, then $p(E)u = 0$, as asserted.

Let p be a polynomial all of whose roots, $\lambda_1, \lambda_2, \dots, \lambda_m$, are simple and nonzero. Corresponding to any root λ_k there is a solution of the difference equation

$$p(E)x = 0$$

Namely, we have the solution $u^{(k)} = [\lambda_k, \lambda_k^2, \lambda_k^3, \dots]$. Let x denote an arbitrary solution of the difference equation. We seek to express x in the form $x = \sum_{k=1}^m a_k u^{(k)}$. Taking the first m components of the sequences in this equation, we obtain

$$x_i = \sum_{k=1}^m a_k \lambda_k^i \qquad (1 \leq i \leq m) \tag{4}$$

The $m \times m$ matrix having elements λ_k^i is nonsingular because its singularity would imply a nontrivial equation

$$\sum_{i=1}^m b_i \lambda_k^i = 0 \qquad \text{or} \qquad \sum_{i=1}^m b_i \lambda_k^{i-1} = 0$$

(This last equation exhibits a polynomial of degree $m - 1$ having m roots.) Equation (4) thus determines $a_1, a_2, \dots, a_m$ uniquely. It remains to be proven that Equation (4) is valid for all values of i . Put $z = x - \sum_{k=1}^m a_k u^{(k)}$. Then $p(E)z = 0$ or equivalently $\sum_{i=0}^m c_i z_{n+i} = 0$ for all n . In other words,

$$z_{n+m} = -c_m^{-1} (c_0 z_n + c_1 z_{n+1} + \dots + c_{m-1} z_{n+m-1}) \qquad (n \geq 1) \tag{5}$$

Note that $c_m \neq 0$ because the polynomial p has m distinct roots and is therefore of degree m . Since $z_i = 0$ for $i = 1, 2, \dots, m$, Equation (5) used repeatedly shows that

$$z_{m+1} = z_{m+2} = \dots = 0 \tag{6}$$

■

Multiple Roots

There remains the problem of solving a difference equation $p(E)x = 0$ when p has multiple roots. Define $x(\lambda) = [\lambda, \lambda^2, \lambda^3, \dots]$. If p is any polynomial, we have seen that

$$p(E)x(\lambda) = p(\lambda)x(\lambda) \tag{7}$$

A differentiation with respect to λ yields

$$p(E)x'(\lambda) = p'(\lambda)x(\lambda) + p(\lambda)x'(\lambda)$$

If λ is a multiple root of p , then $p(\lambda) = p'(\lambda) = 0$, and Equations (6) and (7) show that $x(\lambda)$ and $x'(\lambda)$ are solutions of the difference equation. Thus, a solution is the sequence $x'(\lambda) = [1, 2\lambda, 3\lambda^2, \dots]$. If $\lambda \neq 0$, it is independent of the solution $x(\lambda)$ because

$$\det \begin{bmatrix} \lambda & \lambda^2 \\ 1 & 2\lambda \end{bmatrix} \neq 0$$

and thus, if the sequences are truncated at the second term, the resulting pair of vectors in $\mathbb{R}^2$ is linearly independent.

By extending this reasoning, one can prove that if λ is a root of p having multiplicity k , then the following sequences are solutions of the difference equation $p(E)x = 0$:

$$\begin{aligned} x(\lambda) &= [\lambda, \lambda^2, \lambda^3, \dots] \\ x'(\lambda) &= [1, 2\lambda, 3\lambda^2, \dots] \\ x''(\lambda) &= [0, 2, 6\lambda, \dots] \\ &\vdots \\ x^{(k-1)}(\lambda) &= \frac{d^{(k-1)}}{d\lambda^{k-1}} [\lambda, \lambda^2, \lambda^3, \dots] \end{aligned}$$

THEOREM 2 Theorem on Basis for Null Space

Let p be a polynomial satisfying $p(0) \neq 0$. Then a basis for the null space of $p(E)$ is obtained as follows: With each root λ of p having multiplicity k , associate the k basic solutions $x(\lambda), x'(\lambda), \dots, x^{(k-1)}(\lambda)$, where $x(\lambda) = [\lambda, \lambda^2, \lambda^3, \dots]$.

EXAMPLE 1 Determine the general solution of this difference equation:

$$4x_n + 7x_{n-1} + 2x_{n-2} - x_{n-3} = 0$$

Solution The given equation is of the form $p(E)x = 0$, where $p(\lambda) = 4\lambda^3 + 7\lambda^2 + 2\lambda - 1$. The factors of p are $(\lambda + 1)^2$ and $(4\lambda - 1)$. Thus, p has a double root at -1 and a simple root at $\frac{1}{4}$. The basic solutions are

$$\begin{aligned} x(-1) &= [-1, 1, -1, 1, \dots] \\ x'(-1) &= [1, -2, 3, -4, \dots] \\ x(\tfrac{1}{4}) &= [\tfrac{1}{4}, \tfrac{1}{16}, \tfrac{1}{64}, \dots] \end{aligned}$$

The general solution is

$$x = \alpha x(-1) + \beta x'(-1) + \gamma x(\tfrac{1}{4})$$

or

$$x_n = \alpha(-1)^n + \beta n(-1)^{n-1} + \gamma(\tfrac{1}{4})^n n \quad \blacksquare$$

Stable Difference Equations

An element $x = [x_1, x_2, \dots]$ of V is said to be **bounded** if there is a constant c such that $|x_n| \leq c$ for all n . In other words, $\sup_n |x_n| < \infty$. A difference equation of the form $p(E)x = 0$ is said to be **stable** if all of its solutions are bounded. The difference equation (2) is not stable since one of its solutions is given by $x_n = 2^n$. (Conditioning and stability discussed elsewhere are not related to the concept of a stable difference equation.)

We now ask whether there is an easy method of identifying a stable difference equation.

THEOREM 3 Theorem on Stable Difference Equations

For a polynomial p satisfying $p(0) \neq 0$, these properties are equivalent:

1. The difference equation $p(E)x = 0$ is stable.
2. All roots of p satisfy $|z| \leq 1$, and all multiple roots satisfy $|z| < 1$.

Proof Assume that Property 2 is true of p . Let λ be a root of p . Then one solution of the corresponding difference equation is $x(\lambda) = [\lambda, \lambda^2, \lambda^3, \dots]$. Since $|\lambda| \leq 1$, this sequence is bounded. If λ is a multiple root, then one or more of $x'(\lambda), x''(\lambda), \dots$ will also be a solution of the difference equation. In this case, $|\lambda| < 1$, by Property 2. Since, by elementary calculus (L'Hôpital's Rule)

$$\lim_{n \rightarrow \infty} n^k \lambda^n = 0 \quad (k \geq 0)$$

we see that each sequence $x'(\lambda), x''(\lambda), \dots$ is bounded. (See Problems 1.3.22–23, p. 35.)

For the converse, suppose that Property 2 is false. If p has a root λ satisfying $|\lambda| > 1$, then the sequence $x(\lambda)$ is unbounded. If p has a multiple root λ satisfying $|\lambda| \geq 1$, then $x'(\lambda)$ is unbounded since its general term satisfies the inequality

$$|x_n| = |n\lambda^{n-1}| = n|\lambda|^{n-1} \geq n \quad \blacksquare$$

EXAMPLE 2 Determine whether this difference equation is stable:

$$4x_n + 7x_{n-1} + 2x_{n-2} - x_{n-3} = 0$$

Solution The given equation is of the form $p(E)x = 0$, where $p(\lambda) = 4\lambda^3 + 7\lambda^2 + 2\lambda - 1$. By the preceding example, p has a double root at -1 and a simple root at $\frac{1}{4}$. The equation is therefore **unstable**. ■

An example of a difference equation that has *nonconstant* coefficients arises in the theory of Bessel functions. The **Bessel functions** J_n are defined by the formula

$$J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(x \sin \theta - n\theta) d\theta$$

It is obvious from the definition that $|J_n(x)| \leq 1$. Not so obvious, but true, is the recurrence formula

$$J_n(x) = 2(n-1)x^{-1}J_{n-1}(x) - J_{n-2}(x)$$

If (for a certain x) we know $J_0(x)$ and $J_1(x)$, then the recurrence relation can be used to compute $J_2(x)$, $J_3(x)$, $\dots$, $J_n(x)$. This procedure becomes unstable and useless when $2n > |x|$ because roundoff errors that inevitably occur will be multiplied by the factor $2nx^{-1}$. This factor eventually becomes very large. (See Computer Problem 1.3.2, p. 36.)

For further information on computing functions by means of recurrence relations, see Abramowitz and Stegun [1964, p. xiii], Cash [1979], Gautschi [1961, 1967, 1975], and Wimp [1984].

PROBLEMS 1.3

1. For the sequences following Equation (2), express the first as a linear combination of the second and third.
2. Let p be a polynomial of degree m . Is the solution space of $p(E)x = 0$ necessarily of dimension m ?
3. Let p be a polynomial of degree m , with $p(0) \neq 0$. Prove that if a sequence x contains m consecutive zeros and $p(E)x = 0$, then $x = 0$.
4. Is the operator E **injective (one to one)**? Does it have a right or left inverse? Is it **surjective (onto)**? Define an operator F by $(Fx)_n = x_{n-1}$, with $(Fx)_1 = 0$, and answer the same questions for F . Explore the relationship between E and F . Suppose that V were redefined as the space of all functions on the set $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$, and suppose that F were defined simply by $(Fx)_n = x_{n-1}$. How are the answers to the previous questions affected?
5. What are the eigenvalues and eigenvectors of the operator E ?
6. Consider the infinite series $\sum_{n=1}^\infty x_n v^{(n)}$. What can you say about the question of convergence? Prove that $x = \sum_{n=1}^\infty x_n v^{(n)}$ in the pointwise sense.
7. If $\{v^{(1)}, v^{(2)}, \dots\}$ is adopted as a basis for V , show that $\sum_{i=0}^m c_i E^i$ can be represented by an infinite matrix.
8. (Continuation) Prove that any two operators of the form described in the preceding problem commute with each other.
9. Prove that if L_1 and L_2 are linear combinations of powers of E , and if $L_1 x = 0$, then $L_1 L_2 x = 0$.
10. Develop a complete theory for the difference equation $E^r x = 0$.
11. Give bases consisting of real sequences for each solution space.
 - a. $(4E^0 - 3E^2 + E^3)x = 0$
 - b. $(3E^0 - 2E + E^2)x = 0$

- c. $(2E^6 - 9E^5 + 12E^4 - 4E^3)x = 0$
d. $(\pi E^2 - \sqrt{2}E + \log 2 \cdot E^0)x = 0$
12. Prove that if p is a polynomial with real coefficients, and if $z \equiv [z_1, z_2, \dots]$ is a (complex) solution of $p(E)z = 0$, then the conjugate of z , the real part of z , and the imaginary part of z are also solutions.
13. Solve.
- a. $x_{n+1} - nx_n = 0$
b. $x_{n+1} - x_n = n$
c. $x_{n+1} - x_n = 2$
14. Define an operator Δ by putting

$$\Delta x = [x_2 - x_1, x_3 - x_2, x_4 - x_3, \dots]$$

Show that $E = I + \Delta$. Show that if p is a polynomial, then

$$p(E) = p(I) + p'(I)\Delta + \frac{1}{2}p''(I)\Delta^2 + \frac{1}{3!}p'''(I)\Delta^3 + \dots + \frac{1}{m!}p^{(m)}(I)\Delta^m$$

15. (Continuation) Prove that if $x = [\lambda, \lambda^2, \lambda^3, \dots]$ and p is a polynomial, then $p(\Delta)x = p(\lambda - 1)x$. Describe how to solve a difference equation written in the form $p(\Delta)x = 0$.
16. (Continuation) Show that

$$\Delta^n = (-1)^n \left[E^0 - nE + \frac{1}{2}n(n-1)E^2 - \frac{1}{3!}n(n-1)(n-2)E^3 + \dots + (-1)^n E^n \right]$$

17. Give a complete proof of Theorem 2.
18. Let p be a polynomial such that $p(0) = 0$. Describe the null space of $p(E)$.
19. For $\lambda \in \mathbb{C}$, define $x(\lambda) = [\lambda, \lambda^2, \lambda^3, \dots]$. Prove that if $\lambda_1, \lambda_2, \dots, \lambda_m$ are distinct nonzero complex numbers, then $\{x(\lambda_1), x(\lambda_2), \dots, x(\lambda_m)\}$ is a linearly independent set in V .
20. Prove that if λ is a nonzero root of p having multiplicity k , then the equation $p(E)x = 0$ has solutions $u^{(1)}, u^{(2)}, \dots, u^{(k)}$ in which $u_n^{(j)} = n^{j-1}\lambda^n$.
21. Prove that if $\mu \in (0, \infty)$ and $|\lambda| < 1$, then $\lim_{n \rightarrow \infty} n^\mu \lambda^n = 0$.
22. Prove in detail that a convergent sequence is bounded.
23. Prove Theorem 3 without the hypothesis that $p(0) \neq 0$.
24. Define a sequence inductively by the equation $x_{n+1} = x_n + x_n^{-1}$, where $x_0 > 0$. Determine the behavior of x_n as $n \rightarrow \infty$.
25. Determine whether the difference equation $x_n = x_{n-1} + x_{n-2}$ is stable.
26. Prove that if x is a solution of the difference equation $p(E)x = 0$, then so is Ex .
27. Consider the recurrence relation $x_n = 2(x_{n-1} + x_{n-2})$. Show that the general solution is $z_n = \alpha(1 + \sqrt{3})^n + \beta(1 - \sqrt{3})^n$. Show that the solution with starting values $x_1 = 1$ and $x_2 = 1 - \sqrt{3}$ corresponds to $\alpha = 0$ and $\beta = (1 - \sqrt{3})^{-1}$.

COMPUTER
PROBLEMS 1.3

1. Consider the difference equation $x_{n+2} - 2x_{n+1} - 2x_n = 0$. One of its solutions is $x_n = (1 - \sqrt{3})^{n-1}$. This solution oscillates in sign and converges to 0. Compute and print out the first 100 terms of this sequence by use of the equation $x_{n+2} = 2(x_{n+1} + x_n)$ starting with $x_1 = 1$ and $x_2 = 1 - \sqrt{3}$. Explain the curious phenomenon that occurs.

- 2. Starting with $J_0(1) = 0.76519\,76866$ and $J_1(1) = 0.44005\,05857$, use the recurrence formula in the text to compute $J_2(1), J_3(1), \dots, J_{20}(1)$. What evidence is there that the values are in gross error?
- 3. Consider the difference equation $4x_{n+2} - 8x_{n+1} + 3x_n = 0$. Determine whether it is stable. Find the general solution. Assuming that $x_0 = 0$ and $x_1 = -2$, compute x_{100} using the most efficient method.
- 4. Compute the particular solution in Problem 1.3.27 (p. 35) in the following three ways. Compute these solutions for $1 \leq n \leq 100$ and compare them.
 - a. x_n directly from the recurrence relation
 - b. $y_n = \beta(1 - \sqrt{3})^n$
 - c. $z_n = \alpha(1 + \sqrt{3})^n + \beta(1 - \sqrt{3})^n$, where α is chosen as the unit roundoff error for your computer
- 5. Solve the difference equation $x_{n+2} - (\pi + \pi^{-1})x_{n+1} + x_n = 0$ numerically with $x_0 = 1$ and $x_1 = \pi$. Compute 49 terms and determine the relative error in x_{50} . Then do the same with x_1 changed to π^{-1} and explain the difference in the relative errors of the two cases.